

Tim Martin

Robotics AI Engineer

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Profile

Robotics AI engineer focused on robot learning: teleoperation and data pipelines, fine-tuning and deploying vision-language-action models, imitation and reinforcement learning on humanoids, quadrupeds and robot arms. Background from end-to-end autonomous driving at Bosch to mobile manipulation at ESA.

Experience

Accenture, Next Gen Robotics, Robotics AI Engineer — Oct 2025 – present, 1 year 1 month

- Built teleoperation setups (GELLO) and data pipelines for the Unitree G1 humanoid, a dual-arm YAM setup, Franka and UR arms
- Fine-tuned and deployed VLA (vision-language-action) models on Unitree humanoids, YAM, Franka, UR and OpenArm arms
- Built an agent harness that lets frontier LLMs control robots
- Isaac Sim, Isaac Lab and ROS 2 development
- Boston Dynamics Spot inspection missions, using frontier LLMs for image-based inspection

KIT, Institute for Information Management in Engineering, Research Assistant — June 2025 – Sept 2025, 4 months

- Developed locomotion policies for Unitree Go2 quadrupeds with reinforcement learning (PPO) and adversarial motion priors (AMP)
- Sim-to-real transfer of state-of-the-art control policies, trained in Isaac Lab
- Literature research on state-of-the-art legged locomotion, in close collaboration with a PhD researcher

Bosch Center for Artificial Intelligence, Internship and Master Thesis — May 2024 – Mar 2025, 11 months

Thesis: "Towards End-to-End Autonomous Driving on Non-Academic Data", grade 1.0

- Internship: built a data pipeline to ingest and pre-process large-scale vehicle fleet data for fine-tuning end-to-end driving models
- Thesis: applied the well-known VAD and VADv2 end-to-end driving models to this data and improved them with novel features
- Offline and online evaluation, and deployment in a test vehicle

KIT, Institute for Materials Handling and Logistics Systems, Research Assistant — Dec 2023 – May 2024, 6 months

- Autonomous mobile robots for last-mile delivery
- Perception module: camera calibration and camera-LiDAR alignment
- Some CAD design

ESA, European Astronaut Centre, Internship — Apr 2023 – Oct 2023, 7 months

- Combined a UFactory xArm 5 robot arm with a Clearpath mobile platform into one mobile manipulator
- Implemented a teleoperation framework using Foxglove
- Implemented camera-based autonomous grasping

KIT, Institute for Automation and Applied Computer Science, Research Assistant and Bachelor Thesis — Oct 2021 – Mar 2023, 1 year 6 months

Thesis: "Redispatch 2.0: Forecasting the feed-in quantity of renewable energies using machine learning approaches", grade 1.0

- Co-authored several papers that extend and publish the approach developed in the bachelor thesis

Education

M.Sc. **Karlsruhe Institute of Technology (KIT)**, Mechatronics and Information Technology — Oct 2021 – Sept 2025

- Focus on robotics and AI; final grade 1.1 (top 1%)

B.Sc. **Karlsruhe Institute of Technology (KIT)**, Mechanical Engineering — Oct 2017 – Oct 2021

- Specialization: mechatronics and data science; grade 1.9

Technical Skills

Robot learning: VLA fine-tuning and deployment, RL (PPO, AMP), Sim2Real, LLM agents for robot control

Tools: Python, PyTorch, ROS 2, Isaac Sim, Isaac Lab, LeRobot

Data and hardware: Teleoperation and data pipelines, camera and LiDAR calibration

Robot Platforms

Humanoids and legged: Unitree G1 humanoid, Unitree Go2 quadruped, Boston Dynamics Spot

Arms: Dual-arm YAM setup, Franka, UR5, OpenArm, UFactory xArm 5

Mobile and vehicles: Clearpath mobile base, autonomous test vehicle

Courses

Hugging Face: LeRobot, LLM Course, AI Agents Course, MCP Course

KIT: Python Course

Languages

German: Native

English: Fluent